



Smart Warehouse Grid System

Java-Based Warehouse Management Using Object-Oriented Programming (OOP)

Project Details

Subject: Object Oriented Programming with Java

Project ID: RU-100-01-00012

Student Information

Name: [Aanchal Raj]

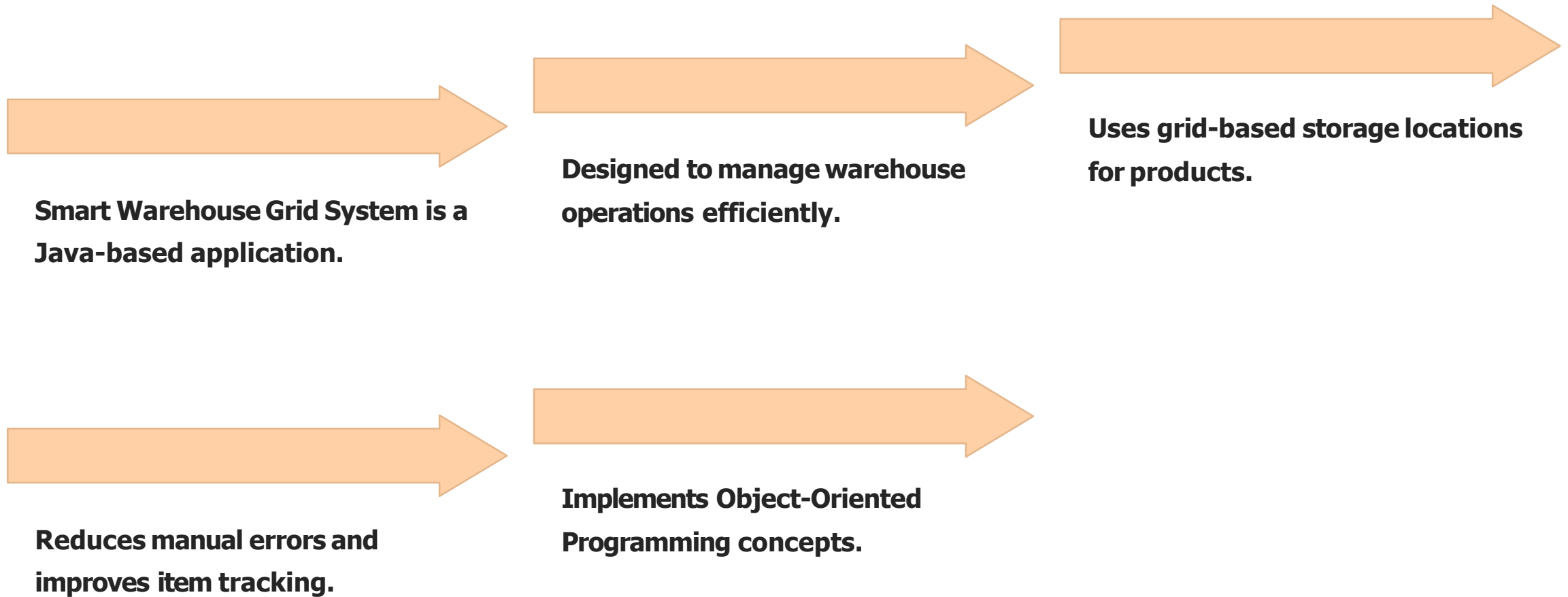
ERP ID: [RU-25-10556]

College: [Rungta International Skill University]

Department: Computer Science s Engineering

Academics Year: 2025-29 (First Year)

Introduction



Project Objectives

Manage warehouse items efficiently.

Store products using grid locations.

Search items quickly using ID or Name.

Update stock information.

Reduce manual work and errors.

Demonstrate OOP concepts such as Encapsulation, Inheritance, Abstraction, and Polymorphism.

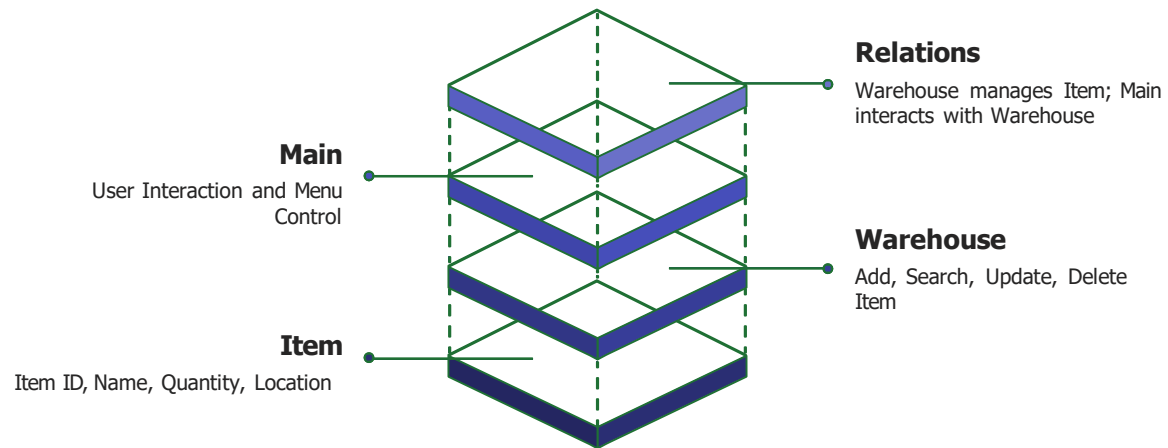
Project Workflow



- **Application starts and displays menu options**
- **User selects from Add, Search, Update, or Delete operations**
- **System validates the input data**
- **Selected operation is processed**
- **Warehouse grid is updated with changes**
- **Results are displayed to the user**
- **User can continue with another operation or exit**

System Design s OOP Implementation

Class Diagram



Classes Used

Item Class:

- **Item ID**
- **Item Name**
- **Quantity**
- **Location**

Warehouse Class:

- **Add Item**
- **Search Item**
- **Update Item**
- **Delete Item**

Main Class:

- **User Interaction**
- **Menu Control**

PROJECT SCREENSHOT

```
1 import java.util.ArrayList;
2 import java.util.Scanner;
3
4 class Item {
5     int id;
6     String name;
7     int quantity;
8     String location;
9
10    Item(int id, String name, int quantity, String location) {
11        this.id = id;
12        this.name = name;
13        this.quantity = quantity;
14        this.location = location;
15    }
16 }
17 public class SmartWarehouse {
18     static ArrayList<Item> warehouse = new ArrayList<>();
19     public static void addItem(int id, String name, int quantity, String location) {
20         warehouse.add(new Item(id, name, quantity, location));
21         System.out.println(x: "Item Added Successfully!");
22     }
23     public static void searchItem(int id) {
24         for (Item item : warehouse) {
25             if (item.id == id) {
26                 System.out.println(x: "Item Found!");
27                 System.out.println("ID: " + item.id);
28                 System.out.println("Name: " + item.name);
29                 System.out.println("Quantity: " + item.quantity);
30                 System.out.println("Location: " + item.location);
31                 return;
32             }
33         }
34         System.out.println(x: "Item Not Found!");
35     }
36     public static void updateItem(int id, int newQuantity) {
37         for (Item item : warehouse) {
```

```
17 public class SmartWarehouse {
18     public static void main(String[] args) {
19         searchItem(id);
20         break;
21
22         case 3:
23             System.out.print(s: "Enter Item ID: ");
24             id = sc.nextInt();
25
26             System.out.print(s: "Enter New Quantity: ");
27             int newQty = sc.nextInt();
28
29             updateItem(id, newQty);
30             break;
31
32         case 4:
33             System.out.print(s: "Enter Item ID: ");
34             id = sc.nextInt();
35
36             deleteItem(id);
37             break;
38
39         case 5:
40             displayAll();
41             break;
42
43         case 6:
44             System.out.println(x: "Thank You!");
45             sc.close();
46             System.exit(status: 0);
47
48         default:
49             System.out.println(x: "Invalid Choice!");
50     }
51 }
```

Results & Advantages

- **Sample Output**

- ◊ Item Added Successfully
- ◊ Item Found at Location (2,3)
- ◊ Item Updated Successfully

Fast item tracking



Better warehouse organization

Reduced human errors



Easy inventory management

User-friendly system



Conclusion s Future Scope

1

Conclusion

- **Successfully demonstrates Java OOP concepts.**
- **Provides efficient warehouse management.**
- **Improves accuracy and productivity.**

1

Future Scope

- **MySQL Database Integration**
- **Barcode Scanner Support**
- **Java Swing GUI**
- **Real-Time Tracking**
- **Web and Mobile Application Version**



CLOSING SLIDE

Thank you